

Bibek Koirala

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SUMMARY

Computer Science Graduate Student (4.0 CGPA) and Graduate Research Assistant specializing in Computer Vision, Deep Learning, and Applied Machine Learning. Experience spans Instance Segmentation, Retrieval-Augmented Generation (RAG) with Large Language Models (LLMs), and full-stack ML pipelines. Background combines academic research, six years of disciplined software engineering and R&D leadership in the Nepali Army, and prior work in healthcare data analytics. First-author paper accepted for oral presentation at IEEE ICMLA 2025.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, SQL

Machine Learning and Deep Learning: PyTorch, TensorFlow, scikit-learn, Transformers, Convolutional Neural Networks (CNN), Vision Transformers (ViT), Mask R-CNN, ResNet, Object Detection, Instance Segmentation, Regression, Support Vector Machines, Decision Trees, Principal Component Analysis (PCA), Kernel Methods, Clustering

NLP and LLMs: Large Language Models (LLM), Vision-Language Models (VLM), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, HuggingFace, FAISS, Sentence Transformers, Vector Databases, Agentic AI

Data and Computer Vision: OpenCV, NumPy, SciPy, Pandas, Matplotlib, Statistics, Convex Optimization

Data Annotation: CVAT (Computer Vision Annotation Tool), Image Annotation, Bounding Box Labeling, Polygon and Mask Annotation, Dataset Curation

Tools and Platforms: Git, GitHub Actions (CI/CD), Docker, Linux, PostgreSQL, VS Code, Jupyter, LaTeX, Experiment Tracking, Model Versioning

EDUCATION

Southern Illinois University Carbondale - M.S., Computer Science Aug 2024 - Present
Graduate Research Assistant, Computer Vision and Object Detection

- CGPA: 4.0 / 4.0
- Coursework:** Machine Learning and Soft Computing, Artificial Intelligence, Deep Learning, Data Mining, Natural Language Processing, Parallel and Distributed Computing, Advanced Database Systems, Cryptography and Network Security

Pulchowk Campus, IOE, Tribhuvan University - B.E., Computer Engineering Nov 2012 - Sep 2016

- Awarded full merit scholarship; ranked in the top 1% of approximately 15,000 entrance exam applicants.
- Coursework:** Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Computer Organization and Architecture, Signal Processing, Big Data, Database Systems, Computer Security, Software Development, Artificial Intelligence

PROFESSIONAL EXPERIENCE

Graduate Research Assistant - Southern Illinois University Carbondale Aug 2024 - Present
Department of Computer Science, Carbondale, IL

- Developed an end-to-end Deep Learning pipeline for instance segmentation on time-stamped microscopy images of bacterial cultures under varying backgrounds, achieving mIoU of 0.91 to 0.98 across growth stages.
- Improved early-stage detection from 3-hour to 2-hour incubation, reaching mAP@0.5 of approximately 0.95; first-author paper accepted for **oral presentation at IEEE ICMLA 2025**.

Army Officer - Software Development and R&D, Nepali Army May 2018 - Dec 2023
Kathmandu, Nepal

- Developed automated artillery fire-control software, reducing targeting error from 50 m to 10 m at 1000 m range and cutting computation latency from minutes (manual) to microseconds through optimized numerical pipelines.

- Designed and developed a Windows desktop dashboard with a relational database backend for managing troop leave and training records, replacing paper workflows and improving administrative throughput.
- Developed decision-support software to compute logistical requirements for Bailey bridge deployment, improving accuracy and field deployment speed under operational constraints.
- Completed multiple leadership and technical training programs emphasizing systems thinking, operational reliability, and disciplined execution under pressure.

Data Analyst - Morang Hospital Pvt. Ltd.

Dec 2016 - Feb 2018

Pathari, Nepal

- Maintained hospital inventory and patient records databases using PostgreSQL; authored queries to track pharmaceutical procurement and sales transactions.
- Performed exploratory data analysis on patient admissions, billing, and inventory datasets to surface operational trends and support data-driven decisions.

PUBLICATIONS

- Koirala, Bibek, et al. "Background-Aware Instance Segmentation for Early Detection of E. coli and Salmonella in Time-Stamped Microscopy Images." *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2025. Accepted for oral presentation. ieeexplore.ieee.org/document/11471447

PROJECTS

Bacteria Detection - github.com/vbek/bacteria-detection

- Developed an end-to-end instance segmentation framework for rapid bacterial detection in microscopy images.
- Achieved significant improvement in early detection capability, reducing effective incubation requirement from 3 hours to 2 hours while maintaining high mAP@0.5 performance.

SEEKBOT: Full-Stack RAG Chatbot - github.com/vbek/SEEKBOT

- Built a 10-node LangGraph pipeline with a local FAISS vector store and HuggingFace Inference API (Llama 3.1 8B and Qwen 2.5 72B) for semantic search across PDF, DOCX, CSV, Markdown, and web content; persistent sessions and fully local embeddings via sentence-transformers.

Plant Disease Detection (Comparative Deep Learning Study) - github.com/vbek/Plant-Disease-Detection

- Compared Convolutional Neural Network (CNN, 34.7M params), fine-tuned ResNet50 (9.9M trainable), vanilla ResNet (36.4M), and Vision Transformer (ViT, 57.5M) on PlantVillage; accuracies of 99.0%, 99.6%, 98.9%, and 97.9% respectively.

News Credibility Detection - github.com/vbek/News-Credibility

- Built an NLP and ML framework classifying news articles using linguistic, semantic, and source-based features; achieved 97% precision and 98% recall on benchmark datasets.

Remittance and Inflation Dynamics - github.com/vbek/Inflation-Dynamics

- Econometric and ML analysis of countries with remittance above 5% of GDP (1960 to 2023), using clustering to examine relationships among macroeconomic indicators.

Crime Victimization Pattern Analysis - github.com/vbek/Victim-Crime-Perception

- Analyzed a sparse, high-dimensional criminology survey to identify victimization patterns and cluster victims by reporting behavior, with results aligning with prior research.

Military Leave Management Dashboard - github.com/vbek/Military-Leave-Management-Dashboard

- Windows-based GUI dashboard with database-backed information system for troop leave and training records, enabling a fully paperless administrative workflow.

LEADERSHIP

- Captain, Pulchowk Campus Cricket Team, 2015 and 2016; competed at the district level in 2016 and 2017.
- Active participant in volleyball, chess, badminton, and running events.
- Six years of military service developed leadership, team coordination, and operational discipline.